Semantic Segmentation Guided RF Micro-Doppler Synthesis and UAV Classification in Low SINR

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***Abstract*—This paper presents a novel approach to synthesize and classify UAV radar micro-Doppler (μD) signatures by utilizing semantic segmentation. The micro-Doppler signatures of UAVs are characterized by Helicopter Motor Rotation (HERM) lines, which appear in parallel to the main radar return and contain information relating to the number of and rotation rate of propellers of the UAV. The rich information in the HERM lines make them of importance when not only classifying but also synthesizing UAV μD signatures. Existing radar μD signature synthesis methods do not explicitly account for the inherent physics of HERM lines, and therefore result in kinematically inaccurate synthetic samples. We propose a novel method that preserves the kinematics of HERM lines by integrating semantic segmentation into the discriminator of a Generative Adversarial Network (GAN). Low-fidelity physics-based simulations of the radar return from UAVs are utilized to pre-train U-Net, thereby enabling the semantic segmentation of HERM lines across a variety of environments with varying noise/clutter levels. Our results show that UAV μD signatures synthesized using the proposed Semantic Segmentation GAN (SS-GAN) enabled improved model training, even in low SINR environments. Furthermore, we propose SS-DNN, which exploits semantic segmentation pretrained with physics-based simulations as a signal conditioner prior to input to a deep neural network, for classification. Results show that the combination of SS-GAN with SS-DNN offers an improvement of 8% for UAV recognition in low SINR.**

***Index Terms*—Semantic Segmentation, Generative Adversarial Networks, Radar, Micro-Doppler, UAVs.**

I. Introduction

The recent boom in accessibility to small unmanned aerial vehicles (UAVs) has led to a proliferation of UAV users from industry to hobbyists. This accessibility has also meant the ease of access for bad faith actors to utilize UAVs, therefore presenting a security risk and urgent need to be able to detect and identify UAV-based threats in a congested airspace. Although a variety of sensors have been proposed to detect UAVs, radar has emerged as a valuable sensor as it can operate remotely and in adverse weather conditions. Optical sensors, such as cameras, are ineffective in many adverse weather conditions, as are acoustic sensors when detecting over large distances or in noisy environments.

Current research on radar-based research for UAV recognition has primarily exploited micro-Doppler (μD) signatures for classification [1], [2]. Micro-Doppler has also been effective in distinguishing UAVs from other small targets, such as birds [3], [4]. A common challenge to the application of deep learning for μD signature classification, however, is a lack of adequate amounts of data for training. The acquisition of real data can be costly and time consuming. Moreover, it is not feasible to acquire data for all possible targets, in various clutter environments, or in operational areas of interest. Thus, not only does the amount of data, but the differences between training data acquired in one environment, while testing in another environment, represents a challenge.

Synthetic data generation has been investigated as a way to improve model training. A variety of methods have been proposed in the literature for UAV signature synthesis. It was shown in [5] that the radar returns of a UAV can be simulated by modeling the radar cross section (RCS) of the blades. Alternatively, [6] simulated the rotor frequencies of UAVs, resulting in realistic μD returns in a noise-less environment. However, each of these models suffer from relying on prior knowledge in the dynamics of UAVs and their radar signatures to simulate returns. These approaches are computationally complex to run; therefore, a thin-wire electromagnetic model was proposed in [7] to simulate noise-less radar returns. More recently, 3D CAD models of drone propellers were utilized to create point clouds for simulating μD returns [8]. However, a common challenge to physics-based simulation methods is that they cannot take into account sensor-specific, target-specific, or environment-specific characteristics, which may also vary with time. For example, a recent study showed that the movement mechanism of the UAV blades affected the resulting μD signature, so that exploitation of mechanical control information could boost classification accuracy [9].

Generative Adversarial Networks (GANs) are an alternative way to synthesize data from a few samples of real data, which can capture such dynamic factors. However, it has been shown that GANs can synthesize kinematically impossible target signatures, which can actually degrade classification accuracy [10]. Integration of μD envelopes as an additional input to a secondary branch in the discriminator has been shown [11] to be effective in reducing kinematic errors in synthetic samples; however, this method is ineffective for UAV signatures because the HERM lines are not an envelope and are not accurately represented in the resulting synthetic samples.

This paper proposes a novel physics-based approach for UAV signature synthesis and classification, which incorporates a semantic segmentation network trained with physics-based

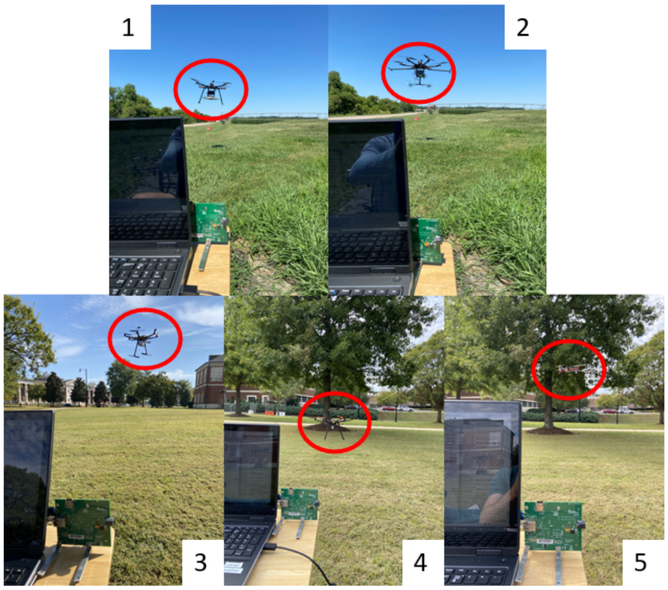


Fig. 1: Collection of radar UAV data on: (1) Quadcopter E, (2) Hexacopter B, (3&4) Hexacopter A, and (5) Quadcopter D.

models into 1) the GAN architecture for μD synthesis and 2) into the classification deep neural network (DNN) for signal conditioning. Although semantic segmentation of range-Doppler, range-Angle maps [12]–[15] and radar point clouds [16] have been explored in the literature, to the best of our knowledge, this work represents the first exploit semantic segmentation in μD signature synthesis and classification.

In Section II of this paper, we first show how differences in environment and SINR impact the visibility of HERM lines in μD signatures and, as result, UAV classification. Section III then describes the low-fidelity simulation method, which animates CAD-based models of the UAVs, to train U-Net [17], a common network used for semantic segmentation in computer vision. Next, we describe 1) the proposed SS-GAN architecture, which integrates Physics-Guided Semantic Segmentation (PhG-SS) into its discriminator, and its use to generate a large amount of kinematically accurate UAV μD data for training; and 2) the proposed SS-DNN, which utilizes PhG-SS as a signal conditioning block prior to its convolutional layers. Our results, detailed in Section IV show that the combination of training the SS-DNN using SS-GAN signatures synthesized from high SINR data results in the highest classification accuracy when testing on UAV signatures with low SINR. Section V concludes by discussing implications for target recognition in new environments from which no training data is available.

II. Impact of Environment on UAV μD Signatures

A. Experiments and Data Collection

To investigate the effect of various clutter environments on UAV μD signatures and their recognition, radar data was collected in three different settings: in an open field, with a tree positioned 10 m from the radar, and with a car in the background 10 m from the radar. Five different UAVs were flown: two hexacopters and three quadcopters. Hexacopters A and B both had propeller blades with lengths of 230mm, and spans of 0.86m and 1.31m, respectively. Quadcopters C, D, and E had propeller blades with lengths of 147mm, 230mm, and 230mm and corresponding spans of 0.24m, 0.55m, and 1.4m, respectively.

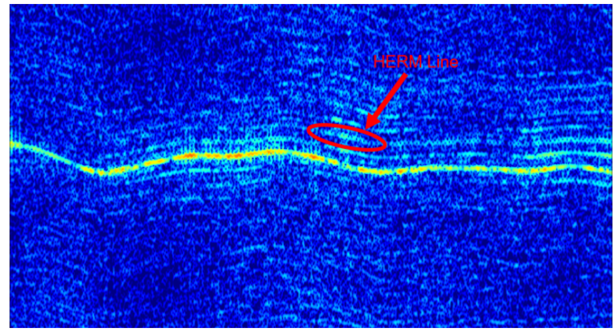


Fig. 2: Hexacopter μD (part of HERM line circled).

A TI AWR2243 single-chip 76-GHz to 81-GHz FMCW radar was operated with a waveform utilizing the full 4-GHz bandwidth and a pulse repetition frequency (PRF) of 6.4-kHz. As shown in Figure 1, the radar was positioned 1m above the ground and operated for increments of 30s per sample. Hexacopter A and quadcopter D were flown approximately 5m from the radar in all three environments, while acquiring ten recordings per UAV. The other three UAVs, hexacopter B and quadcopters C and E, were only flown over an open field, with 8 samples collected per quadcopter and 3 samples of the hexacopter given battery time constraints when flying. This resulted in a total of 79 recordings of 30s of data across all 5 UAVs. The final dataset was obtained by splitting the recordings into 6s segments, yielding a total of 395 samples.

The micro-Doppler signature for each 6s sample was then obtained by computing the spectrogram - the square modulus of the short-time Fourier transform (STFT) with a window length of 256, overlap of 200, and 4096 FFT points. An example of the spectrogram of a UAV radar return can be seen in Figure 2, where Helicopter Rotation Modulation (HERM) lines, which are lines that are spaced between one another corresponding to the rotation rate and number of blades present in the UAV [18], appear parallel to the main micro-Doppler return. This occurs as the size of the window for the STFT is large, whereas a shorter window size can reveal the blade flashes of the UAV. In this work, we focused on signatures with HERM lines as each individual propeller has its own unique HERM line pattern according to its rotation rate, making them a critical factor for discriminating different types of UAVs.

B. Impact on SINR on Feature Distribution

To understand the impact of clutter from various environments on the measured UAV μD signatures and their discriminability, we utilized Principle Component Analysis (PCA) to visualize the feature spaces of the data in two dimensions. Comparing the distribution of samples based on

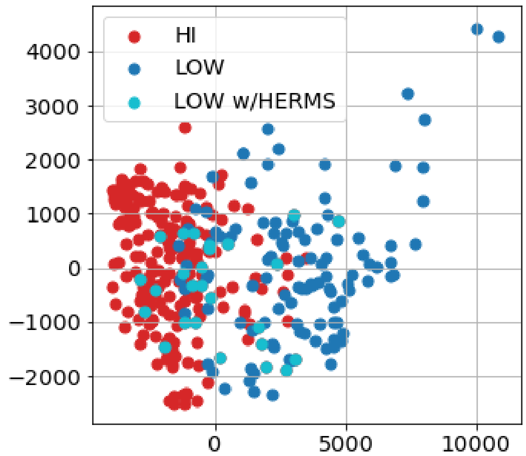


Fig. 3: PCA distribution of real data at High and Low SINR.

the setting in which data was acquired (open field, near tree, near car), we found that the feature distributions closely overlapped. If however, we instead observed the variations in feature space based on signal-to-interference-plus-noise ratio (SINR), we found that significant shifts in the distribution could be observed, as shown in Figure 3.

First, we partition the dataset into "High SINR" and "Low SINR" groups with SINRs of >10 dB and between 0-10 dB, respectively. This resulted in 107 High SINR samples which contained noticeable HERM lines, while among the 226 Low SINR samples, we observed that there were 198 samples that exhibited no visible HERM lines. There is a distinguishable difference in the distribution of the samples with High SINR versus those with Low SINR (but without HERM lines). Although there is some overlap between High SINR samples and those with Low SINR but with visible HERM lines, there remains a noticeable displacement in the centroids of their respective PCA clusters. As we will see quantitatively in later sections, this shift in distribution is the root cause for deep learning models trained in one environment (SINR) being ineffective in generalizing to classification of testing data acquired in a different environment (SINR).

III. Proposed Method

To address the challenge of UAV recognition in Low SINR operational environments, where no training data at that SINR is available in advance, we propose 1) SS-GAN: a novel method for synthesizing training samples that exhibit improved representation of HERM lines, and 2) SS-DNN: a novel DNN architecture for classification that offers improved performance in Low SINR environments. At the core of both networks is PhG-SS - a semantic segmentation network that has been trained with physics-based models of UAV blade rotations, as discussed next.

A. PhG-SS: Physics-Guided Semantic Segmentation

Semantic Segmentation is an computer vision technique that aims to partition an image into smaller segments that correspond to objects within an image. Each pixel of an image is labeled with a corresponding class of what is being represented. In our case, we wish to isolate and identify the pixels within a μD signature that correspond to the full return signal of the UAV, which includes the HERM lines. Essentially, this process is binary semantic segmentation, where each pixel is categorized as "UAV signal return" or "not UAV signal return."

In this work, we chose a popular network for semantic segmentation of digital imagery, U-Net, as the backbone for this task. The key however, is to train the U-Net in such a way that it is guided by physics-based models in its segmentation process. Physics-guided training is essential, especially in Low SINR samples, where the HERM lines may not be readily evident, and is provided by taking advantage of physics-based UAV simulations based on CAD-models of the UAVs.

1) Simulating UAV μD from Animated CAD Models: A low-fidelity model of the radar returns from a UAV is generated by representing the physical structure of the UAV with point clouds extracted from CAD-based models of a quadcopter and hexacopter. These point-clouds were then loaded into MATLAB as three layers: the full UAV, the propellers, and each individual propeller. The full UAV point-cloud was normalized to a $1 \times 1 \times 1$m box, which could then be adjusted in size according to the real UAVs used for this work. To simulate the basic motion of the UAV, each individual propeller was initially animated according to a specified rotations per minute (RPM). The new propeller positions were then mapped back to the full set of propellers, with the new position of all the propellers then mapped back to the full UAV. The entire UAV was then moved according to the specified speed of the UAV, the capture rate of the radar, and a randomly generated circular path. To speed up simulation time and minimize computational complexity, we utilized 10 points per component for the propellers and body of the UAV, thereby preserving critical target features while reducing computational complexity. Combined with a model for the radar backscatter based on the radar range equation, the animated UAV models were used to generate the complex IQ samples of the expected radar return, which were then processed in the same method as the real UAV data.

2) Physics-Guided Training of U-Net: The U-Net model for semantic segmentation is trained with the low-fidelity simulated dataset to semantically segment the main signal return and corresponding HERM lines of UAV μD signatures across high and low clutter samples. For the low-fidelity simulated dataset, corresponding segmentation maps for each sample is needed to then train the U-Net to properly segment the UAV signal from the background clutter. To create the ground truth segmentation maps, each sample without any injected noise is converted into a binary image, with each pixel containing the UAV signal being set to 1 and all others at 0. The dataset was further increased in size with each noise-less sample being injected with noise corresponding to SINRs of 5db, 10db, 15db, and 20db. For all 5 noise levels, the same segmentation map was used for training the U-Net. An example of these samples and their corresponding shared segmentation map can be seen in Figure 5. The U-Net was

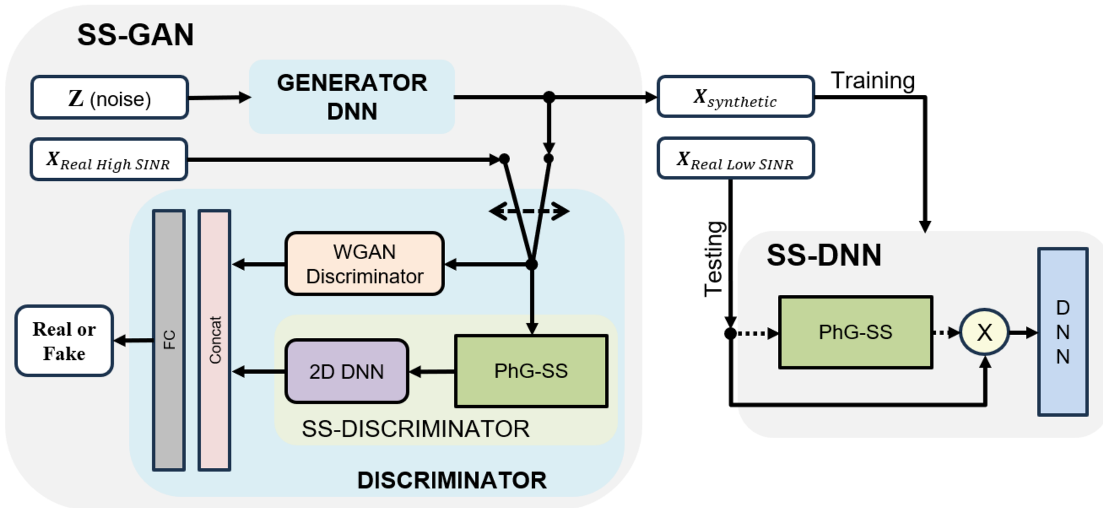
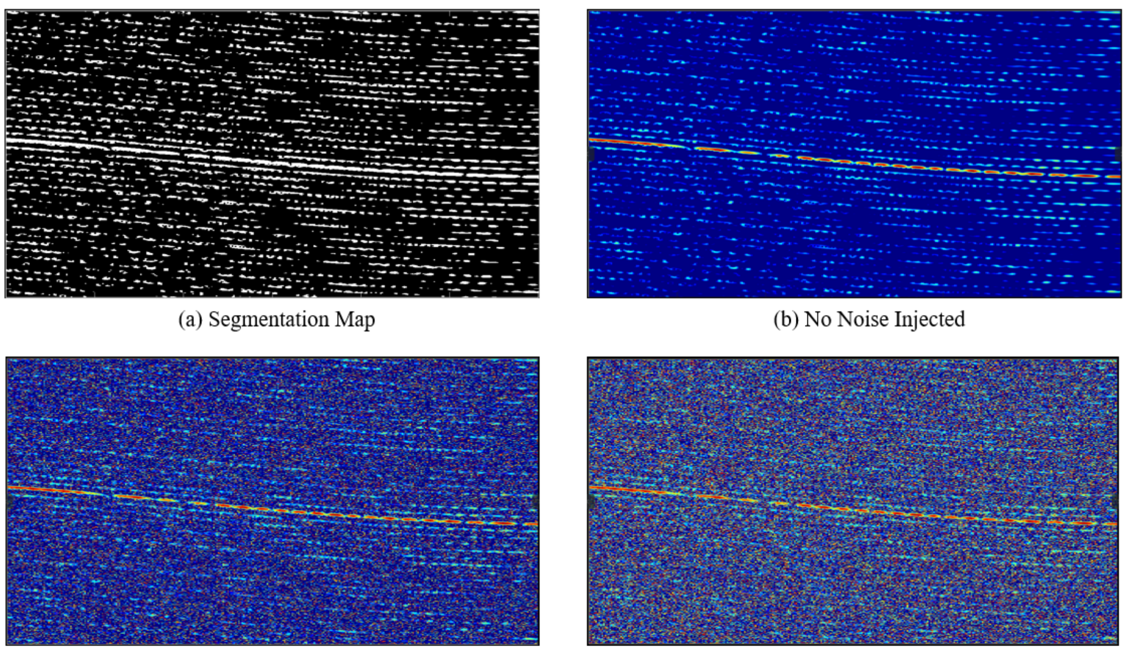


Fig. 4: Overview of proposed approach for physics-guided semantic segmentation-based μD synthesis and classification.



(a) Segmentation Map

(b) No Noise Injected

(c) 20dB SINR

(d) 5dB SINR

Fig. 5: Low-Fidelity Simulated Spectrograms

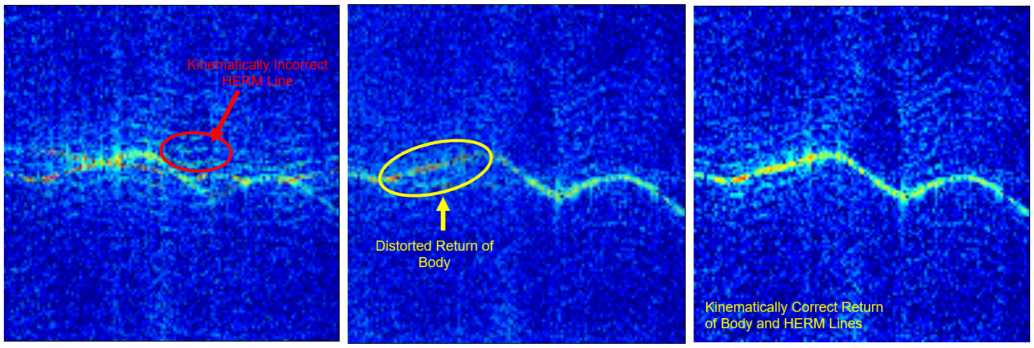
then trained on this full dataset for 40 epochs. Upon training with only low-fidelity simulated samples, the U-Net was then observed to segment the full UAV signal returns on random samples of the real UAV dataset with high SINR.

B. SS-GAN: Semantic Segmentation-based GAN

Physics-guided semantic segmentation is a useful process for informing GANs about whether the samples generated are true to the real data or fake. Consequently, in addition to the main discriminator branch that takes as input the micro-Doppler signature, we add an auxilliary discriminator branch that takes as input the segmentation map provided by PhG-SS. A 2D DNN of 2 convolutional layers then extracts the features from the segmentation map before being concatenated with the output of the main discriminator branch, which is then provided to a fully connected layer. In this way, the inherent physics of the HERM lines is accounted for when determining whether the synthesized sample is real or not. As a result, synthetic samples that exhibit HERM lines with greater kinematic fidelity are generated.

C. SS-DNN: Semantic Segmentation for Signal Conditioning

Physics-guided semantic segmentation can also be utilized to improve the classification of UAVs in high clutter environ-



(a) WGAN (b) MBGAN (c) SS-GAN

Fig. 6: GAN-synthesized hexacopter Spectrograms

ments. As shown in Figure 4, the segmentation map resulting from the PhG-SS block can be used to mask the input μD signature and effectively serve as an image denoiser, supplying the DNN for classification with distinguished HERM lines.

In this work, to exemplify the benefits of the proposed approach, we construct a Convolutional Neural Network (CNN) comprised of three 2D convolutional layers with 32, 64, and 128 filters, respectively. The first two convolutional layers are followed by batch normalization layers, flattened, and input to a dense layer whose units are based on the number of classes - in this case, the number of UAVs.

IV. Results

We evaluate our proposed approach in two ways: first, we analyze the kinematic fidelity of the SS-GAN synthesized μD signatures, comparing the result of that obtained with a conventional Wasserstein GAN (WGAN) and a Multi-Branch GAN (MBGAN), which we proposed for human μD signature synthesis in prior work [11], which takes as an auxilliary input μD signature envelopes. In particular, each GAN was trained using 80% of the real, high SINR UAV data for 25,000 epochs, resulting in the generation of 500 samples per UAV. Second, we conduct an ablation study in which the classification accuracy attained in Low SINR data is compared for two different DNNs, the use of real versus synthetic training data, and presence of distinct versus obscure HERM lines.

A. Kinematic Fidelity

Examples of the UAV μD signatures generated from the WGAN, MBGAN and proposed SS-GAN are shown in Figure 6. It may be observed that the envelope-based MBGAN struggles to maintain the power of the main return of a UAV, while the WGAN creates HERM lines that are not kinematically correct, deviating from the characteristic parallelism observed in real HERM lines. In contrast, SS-GAN results in samples that maintain the strength of the main return and have HERM lines in parallel with the μD returns from the UAV body.

The improvement resulting from training with SS-GAN synthesized samples is quantitatively assessed by training a 3-layer CNN with the data from each GAN and comparing the resulting classification accuracies attained. Note that in all cases, we only use real data with high SINR during both training and testing. This scenario is representative of the case where the UAV of interest is being flown in an environment consistent with that in which the training data was obtained.

The results are shown in Table I, where a small training set of real UAV data can be seen to only achieve a classification accuracy of nearly 67%. Utilizing either the WGAN or MBGAN synthesized datasets for training does show improvements in classification accuracy, but the best performance was achieved when training on data generated by the proposed SS-GAN network, boosting the classification accuracy by nearly 10% as compared to only utilizing the real dataset for training and testing.

B. UAV Recognition in Low SINR

In typical operational scenarios, it is not feasible to obtain training data from the operational environment of interest. We thus consider the challenging scenario of UAV recognition in Low SINR environments, without having any Low SINR data utilized in the training process. In particular, we conduct an ablation study where we compare the accuracies attained from two different DNNs: the 3-layer CNN and the proposed SS-CNN, which uses physics-guided semantic segmentation maps as the input to the same 3-layer CNN. We compare the accuracy attained when using real data only for training with that of utilizing SS-GAN synthesized training data. Note that while the training process uses only high SINR samples, the test data is exclusively drawn from Low SINR samples. Results for Low-SINR samples that possess distinct HERM lines and those in which HERM lines are not visible are separately reported in Table II.

TABLE I: Efficacy of Training with SS-GAN

Network	Training Data	Testing Data	Accuracy
CNN	Real @ High SINR	Real @ High SINR	66.6%
	WGAN - High SINR		75.0%
	MBGAN – High SINR		72.6%
	SS-GAN – High SINR		**76.2%**

TABLE II: Classification of Low SINR UAV Signatures

Network	Training Data	Testing Data	Accuracy
CNN	Real @ High SINR	Real @ Low SINR	41.6%
	SS-GAN - High SINR		28.5%
	Real @ High SINR	Real @ Low SINR w/HERMs	56.2%
	SS-GAN – High SINR		57.7%
SS-CNN	Real @ High SINR	Real @ Low SINR w/HERMs	65.4%
	SS-GAN – High SINR		**73.1%**

It may be observed that when HERM lines are not apparent, the CNN yields very poor classification accuracy regardless of which dataset is utilized for training. When HERM lines are apparent, however, improvements of nearly 15% and 30% are seen when training with the real UAV dataset at high SINR and the high SINR SS-GAN synthesized dataset, respectively. In this case, utilizing the synthesized dataset leads to a slight improvement in classification accuracy of approximately 1.5%.

If, however, the proposed SS-CNN is utilized, the classification accuracy increases by 9%-15% relative to just a CNN alone. The best overall accuracy of 73.1% is attained when the SS-GAN synthesized data is utilized to train the SS-CNN classifier. It should be underscored that this performance was achieved despite using only High SINR data to train the SS-GAN. ***No data at the same (low) SINR of the testing dataset was used during training.*** Also note that the accuracy attained (73.1%) is only 3% lower than that obtained when both the training and testing data have high SINR.

The implications of this result is that the proposed approach is not just effective for classifying UAV signatures in low SINR, but that this technique potentially can mitigate the performance degradation incurred in dynamic clutter environments, where differences in the distribution of training data with testing data harm a DNN's ability to generalize to new environments. The role of the PhG-SS is essentially that of a denoiser, suppressing the variations due to dynamic clutter - so long as the HERM lines are distinct enough that the physics-guided semantic segmentation map provides relevant features for classification.

V. Conclusion

HERM lines represent a rich source of information that aides UAV recognition; however, conventional GANs often struggle to generate accurate kinematic representations of HERM lines and classifiers struggle in Low SINR environments. This paper addresses these challenges by proposing a method to maintain the inherent kinematics of HERM lines using physics-guided semantic segmentation (PhG-SS) attained through pre-training a U-Net model with low fidelity physics-based simulated UAV μD signatures. The PhG-SS block is integrated into the architecture of the proposed SS-GAN network for UAV signature synthesis. The synthesized samples are then used to train an SS-CNN in which the segmentation map of the PhG-SS block is used as a signal conditioner prior to the convolutional layers. Our results show that the combination of SS-GAN and SS-DNN can provide improved recognition accuracy for low SINR samples, even when only samples at high SINR are available for training support. The implications of these results are that the proposed approach can provide a way to improve generalization and target recognition in dynamic environments.

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